

# **Project Proposal for Arrhythmia Treatment Outcome Prediction AI**

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## **Project Background**

Cardiovascular disease is widely recognized as the number one cause of death in the United States. Within this broad category, cardiac arrhythmias contribute to this burden by disrupting the heart's normal rhythm and, in severe cases, leading to stroke, heart failure, or sudden death. The most common type of arrhythmia is atrial fibrillation, which is present in more than 10 million American adults and significantly increases stroke risk. To treat atrial fibrillation, doctors perform catheter ablation by inserting a thin catheter through a blood vessel and scarring areas that produce abnormal signals. However, precise catheter placement is crucial because improper positioning can result in unintended tissue damage and treatment failure. Currently, the overall ablation success rate for atrial fibrillation ranges between 37-51%, so there is a major need for improvement.

## **Proposed Idea**

The goal of this project is to build a system that helps doctors, clinicians, and researchers to verify whether an ablation procedure successfully treats a patient with atrial fibrillation. The system will be composed of two parts: a binary classification model that takes the positions of the catheters during ablation to predict whether the patient will recover or not, and a software application (UI) that allows users to select catheter points from a 2D or 3D representation of the atria and enter experimental settings as input for the model

## **Technical Project and Specifications**

The system consists of two primary components: a machine learning backend that predicts ablation outcome from catheter placement data, and a client-facing software application that allows clinicians to input procedural data and view predictions. These components will communicate through a lightweight API layer.

## *I. Machine Learning Component*

The model pipeline will begin with data preprocessing on the Cleveland Clinic datasets, including handling missing values, normalizing continuous variables such as catheter temperature and pressure, and encoding the catheter-position variables. Feature selection techniques will be used to identify which catheter positions and experimental parameters are most predictive of ablation success. Multiple model architectures will be trained and compared, including decision trees, gradient boosting models, and neural network classifiers, with the best-performing model selected through cross-validation and hyperparameter tuning, then serialized for deployment.

## *II. Data*

The data will primarily consist of binary variables indicating whether a catheter was placed at a specific location within the heart, along with experimental metadata such as catheter temperature and pressure, the type of ablation procedure performed, and other relevant parameters. The Cleveland Clinic will provide datasets for paroxysmal and persistent atrial fibrillation, with additional datasets for other arrhythmias potentially included if time permits.

## *III. Backend*

The backend will be built using FastAPI. It will expose an API endpoint that accepts catheter placement data and experimental metadata as input, loads the trained model, and returns a prediction along with a confidence score. FastAPI provides built-in request validation and automatic API documentation, which will make it easier to keep the frontend and ML components in sync.

## *IV. Frontend / UI*

The frontend will provide an interactive 2D representation of the atria, allowing users to click to mark catheter placement points rather than manually entering coordinates. It will also include form fields for experimental metadata, such as catheter temperature, pressure, and ablation type, and a results view displaying the predicted outcome and confidence score returned by the backend.

## **Related Work**

<https://www.jnimedtech.com/en-US/news-events/biosense-webster-launches-new-version-carto-3-electro-anatomical-mapping-system> - **CARTO 3 System 8**

A 3d heart mapping system used in cardiac ablation procedures. It visualises catheter locations and uses ML algorithms to automatically left atrial anatomy. This provides strong inspiration for the catheter selection UI and interactive atrial visualization.

<https://www.medtronic.com/en-us/healthcare-professionals/products/surgical-energy/ablation/mapping-ablation-products/affera-mapping-system.html> - **Affera mapping system**

A cardiac mapping and ablation platform that creates high-fidelity cardiac anatomy, visualizes catheters in real time, and automatically generates lesion/location tags. We can use this for a variety of inspirations, like results interface and catheter point presentation.

<https://www.cardiovascular.abbott/us/en/hcp/products/electrophysiology/mapping-systems/ensite-x.html> - **EnSite X EP System**

A cardiac mapping system that created electro-anatomical maps for use during procedures. We can reference this for our catheter position visualization.

## **Work to be done**

### *Machine Learning*

- Preprocess and analyze Cleveland Clinic data
- Select, train, compare, and evaluate classification models
- Tune and save final models for deployment

### *Backend*

- Build FastAPI endpoint
- Connect the trained models to the API
- Validate inputs and return outcome and confidence score

### *Frontend/UI*

- Build interactive 2D atria interface for catheter placement
- Add fields for temperature, pressure, ablation type, and other inputs
- Display prediction results and confidence

### *Integration & Testing*

- Connect frontend, backend, and ML models
- Test the complete system and fix any issues

## **Management Plan**

To successfully complete this project, we will do multiple things. First, we will meet as a team every Friday at 10:30 am at KSL to ensure we are up to date with the work that needs to be done. We will communicate using a text message group chat and it is expected that everyone will respond in a timely manner. All code will be tracked, stored, and edited using a GitHub repository. All other documents will be stored in a shared Google Drive. The work for the project will be broken into 2 main sections, machine learning and app development. Jason and Chris will work on the machine learning development and Devak and Lucas will work on the app development. All team members will split the work for deliverables evenly.

### Timeline

| Event                                         | Date               |
|-----------------------------------------------|--------------------|
| Meet team & decide on project                 | Week 2             |
| Submit Project Proposal                       | Week 3 (Sept 10th) |
| Receive and process Cleveland Clinic data     | Week 4             |
| Backend API setup                             | Week 5             |
| Initial ML models trained                     | Week 6             |
| Initial UI & 2D atria interface built         | Week 7             |
| Initial integration and testing               | Week 8             |
| Submit Progress Report + Demo                 | Week 9 (Oct 22nd)  |
| Compare and select ML models                  | Week 10            |
| Frontend takes in all required inputs         | Week 11            |
| Integration and testing                       | Week 12            |
| Finalize code & start to work on deliverables | Week 13            |
| Work on deliverables                          | Week 14            |
| Submit Final Project                          | Week 15 (Dec 3rd)  |
| Present at Intersections                      | Week 15 (Dec 4th)  |

**References:**

- [Restoring the Rhythm: Atrial Fibrillation Diagnosis, Treatment, and Prevention - Clinical Advisor](#)